Conceptual Design Document — Rev 00-R Full

Gen-3 Safe Helicopter / Heavy Modular Amphibious VTOL Platform

Realistic, Defensible & Manufacturable Investor Baseline
Gen-3A 100-ton Pure Rotorcraft | Gen-3B 200-ton Hybrid Buoyant Rotorcraft

Founder:	Amirbahador Atai
Date / Revision:	2026 | 00-R Full (Realistic Safety-Centered Baseline)
Status:	Concept Baseline / Pre-Feasibility — Full Investor & Technical Edition
Classification:	Proprietary Concept Information
Pages:	16

1. LEGAL NOTICE AND CAVEATS

All numerical values are conceptual design targets requiring validation by CFD, FEA, aeroelastic analysis, system safety assessment, testing and certification authorities. This document does not claim certified safety, airworthiness, manufacturability or guaranteed performance. Cost and schedule figures are order-of-magnitude estimates only. This document is not an offer, prospectus or investment recommendation. Recipients must conduct independent technical and financial due diligence.

2. EXECUTIVE SUMMARY FOR INVESTORS

Gen-3 is a family of heavy amphibious VTOL aircraft built around one radical safety idea: every occupant sits inside a centrally located, rapidly detachable, independently buoyant capsule that carries its own underbody ducted fans and multi-canopy parachute system. In an emergency the capsule separates from the rest of the aircraft in 5–7 seconds and becomes a self-contained survival craft.

Gen-3A (100 t) is a pure mechanical-lift rotorcraft and the nearer-term product. **Gen-3B (200 t)** is a hybrid that uses helium buoyancy to offset 50–65 % of its weight, keeping installed mechanical power within near-term technology. Both share the same capsule philosophy, modular architecture and amphibious hull.

This Rev 00-R Full baseline deliberately reduces earlier over-optimism. Mechanical T/W is set at ≈ 1.45–1.50, separation at 5–7 s, fan counts are modest, and 15–20 % development margins are retained. The result is a concept that remains ambitious yet is more defensible under technical and financial due diligence, and more compatible with manufacturability and certification realities.

3. VALUE PROPOSITION AND MARKET CONTEXT

3.1 Core Safety Differentiator

Conventional heavy helicopters permanently attach occupants to the rotors, engines and fuel. In a catastrophic failure the entire aircraft is the survival cell. Gen-3 inverts that logic: occupants leave the failing aircraft inside a purpose-designed capsule with independent lift, flotation, life support and distress systems. This is the strongest commercial and technical selling point of the concept.

3.2 Market Need

Heavy vertical lift with genuine water operability remains underserved. Existing platforms (CH-47, CH-53K, Mi-26, AW101 class) offer limited water crashworthiness and no independent occupant escape system. Offshore energy, disaster response, Arctic/remote logistics and certain special-operations roles accept high residual risk because no better alternative exists. A credible safety step-change in this segment has pricing power and potential regulatory support.

3.3 Why Two Variants

Gen-3A is the nearer-term, lower-risk product with a clearer certification path. Gen-3B is the longer-term capacity step-up that re-uses capsule, control and modular technology. Shared investment in the common safety architecture improves capital efficiency.

4. DESIGN PHILOSOPHY — REALISTIC BASELINE

Safety first, then feasibility, then performance. Earlier iterations targeted T/W = 1.7 and separation under 4 s. Those targets have been relaxed to levels that still deliver a decisive safety advantage while remaining within credible near-term technology:

- Mechanical T/W for Gen-3A ≈ 1.45–1.50 (healthy hover margin, lower power and structural load).
- Capsule separation timeline 5–7 seconds (still extremely fast, more achievable than 3.5 s).
- Modest fan counts to control interference, structural mass and certification complexity.
- Simplified parachute systems (6–8 canopies) to reduce entanglement risk.
- Installed power with explicit battery buffer and conservative turbogenerator assumptions.
- 15–20 % development margins retained in mass and power.
- Manufacturing approach based on modular line-replaceable units and existing industrial processes.

5. COMMON REQUIREMENTS (BOTH VARIANTS)

5.1 Central Safety Capsule

Centrally located, rapidly detachable, independently buoyant capsule for all occupants. Capacity: 24–30 pax + 2 crew (Gen-3A) or 28–36 pax + 2 crew (Gen-3B). Features:

- 72-hour independent life support and emergency oxygen with positive-pressure capability.
- GNSS + satellite distress beacons and dedicated flotation volumes.
- Underbody ducted fans for post-separation attitude control and descent-rate management.
- Multi-canopy parachute system (sequenced, reefed, modular) — 6–8 canopies.
- Target full separation time: 5–7 seconds after decision.
- Independent battery power for all post-separation functions.
- Crashworthy seats and energy-absorbing floor sized for flight and water impact.

5.2 Amphibious Capability

Controlled water landing, flotation and takeoff in Sea State 3–4. Trimaran/catamaran-inspired hull with crushable keel and energy-absorbing underside. Water-ingestion risk into underbody fans is a primary design and test driver.

5.3 Structural and Fuel Safety

Primary structure 60–70 % carbon-fibre composite by mass. Metallic joints, lightning protection and impact structures retained. Fuel only in self-sealing crashworthy central bladders; no fuel in peripheral modules. Automatic high-voltage isolation on water contact.

5.4 Modularity and Fire Response

Capsule, fan arms and turbogenerator packs are discrete modules with standardised interfaces. Fire sequence: Detect → Isolate → Suppress → Stabilize → Jettison (only if necessary, under flight-control authority with verified clear trajectory).

6. DESIGN CRITERIA, FORMULAS AND EXTENDED ASSUMPTIONS

$T_{target} \approx 1.45\text{–}1.50 \times W$ (Gen-3A mechanical)
$A_{disk} = \pi (D/2)^2$
$P_i = T^{3/2} / \sqrt{(2 \rho A)}$ (ideal induced power)
$L_b = (\rho_{air} - \rho_{He})\, g\, V$ (gross buoyancy)

6.1 Extended Design Assumptions

- Sea-level International Standard Atmosphere ($\rho \approx 1.225$ kg/m³); no ground-effect credit in baseline hover power.
- Fan figure of merit and propulsive efficiency consistent with current large ducted-fan / open-rotor technology at the chosen disk loadings.
- Turbogenerator specific power 0.45–0.65 kW/kg class including gearbox and accessories (subject to supplier data).
- Battery specific energy 180–230 Wh/kg (near-term technology) for capsule emergency storage.
- Composite structural mass fraction 60–70 % achievable with out-of-autoclave and automated fibre-placement processes.
- Helium purity, containment leakage and inspection intervals within current large-airship industry practice.
- Lightning protection, bonding and static discharge follow established aerospace and airship standards.
- No exotic materials or unproven manufacturing processes required for the baseline concept.
- 15–20 % development margins applied to mass and installed power.
- Control architecture at least triple-redundant with an independent high-integrity separation path.

7. GEN-3A — 100-TON PURE ROTORCRAFT (DETAILED)

7.1 Mission Role

Nearer-term product for heavy logistics, disaster response, offshore support and special-operations insertion where runway independence and water operability are required. Emphasis on hover performance, safety and flexibility rather than high cruise speed.

7.2 Key Performance Targets

Parameter	Value
Operational Mass	100 t
Target Mechanical Thrust (T/W ≈ 1.45–1.50)	145–150 tf (≈ 1.42–1.47 MN)
Hover Power (Baseline / Design with margin)	~24–28 MW / 36–42 MW
Installed Generation	55–65 MW (6–8 × ~8–9 MW + battery buffer)
Lift Distribution	6–8 peripheral fans (11–12 m dia.) + 4 capsule underbody fans (5–6 m)
Approx. Total Disk Area	~900–1 100 m²
Approx. Disk Loading	~130–165 kg/m²
Capsule Separation Time (target)	5–7 seconds
Parachute System	6–8 main canopies (modular, sequenced)
Crew + Passengers	2 + 24–30
Fuel (protected central only)	5–6 t
Development Mass / Power Margin	15–20 %

7.3 Preliminary Mass Ledger

Subsystem	Mass (t)	Notes
Primary structure (composite + metallic)	26–29	60–70 % composite target
Peripheral arms, fans, ducts	14–17	6–8 modules
Motors / inverters	8–10	Distributed electric
Turbogenerators + fuel system	12–15	Incl. mounts & plumbing
Safety capsule (complete)	11–13	Structure, LS, fans, chutes, battery
Avionics / flight controls	4–5	Triple-redundant
Landing gear + flotation	5–6	
Fuel	5–6	Central bladders only
Occupants + emergency stores	4–5	
Target close	≈ **100 t**	With 15–20 % development margin

7.4 Power and Energy Architecture

Six to eight turbogenerators in the 8–9 MW class. Battery buffer of approximately 12–18 MWh provides peak smoothing, engine-out power and the entire post-separation energy need of the capsule. Fuel is stored only in protected central self-sealing bladders. Electrical architecture is high-voltage DC with multiple bus segments and automatic isolation on fault or water contact.

7.5 Flight Control and Redundancy

Fly-by-wire with at least triple redundancy. Control allocation handles multi-fan layout and reconfiguration after loss of one or more fans. Capsule separation is a high-integrity function with independent power and sensing, not reliant on the primary flight-control computers.

7.6 Why Gen-3A Is More Credible Than Earlier Baselines

- Lower T/W (1.45–1.50) reduces installed power and structural loads versus earlier 1.7 target.
- Fewer fans reduce interference, control complexity and structural mass.
- 5–7 s separation window remains extremely fast but gives realistic sequencing margin.
- 6–8 canopies lower entanglement probability versus earlier 12-canopy concepts.
- Mass ledger written to close at 100 t with explicit 15–20 % development margin.
- Manufacturing approach based on modular LRUs and existing industrial processes.

8. GEN-3B — 200-TON HYBRID BUOYANT ROTORCRAFT (DETAILED)

8.1 Mission Role

Longer-term, higher-capacity platform for ultra-heavy logistics, persistent maritime presence and large-scale disaster response. Buoyancy keeps residual mechanical power demand inside the envelope of available turbogenerator technology.

8.2 Key Performance Targets

Parameter	Value
Operational Mass	200 t
Helium Volume (target)	180 000 – 220 000 m³
Gross Buoyancy (approx.)	105 – 130 t equivalent
Mechanical Thrust Requirement	~90–120 tf ($T/W_{mech} \approx 0.45$–0.60)
Hover Power (Baseline / Design)	~30–40 MW / 50–65 MW
Installed Generation	70–90 MW
Lift Distribution	8–10 peripheral fans (11–12 m) + 4–6 capsule underbody fans (5.5–6.5 m)
Capsule Separation Time (target)	5–7 seconds
Parachute System	8 main canopies (modular, sequenced)
Crew + Passengers	2 + 28–36
Development Margin	15–20 %

8.3 Buoyancy Architecture and Residual Risk

Helium volume is sized so that static buoyancy offsets roughly 50–65 % of operational mass. Residual mechanical thrust demand therefore falls into a range that can be met with a more modest fan count and installed power than a pure 200 t rotorcraft would require. The envelope is multi-cell with lightning protection and continuous leak detection. Integration geometry must preserve a short, clear capsule separation path.

Even with realistic buoyancy credit, Gen-3B remains higher-risk and longer-timeline than Gen-3A. Helium containment mass, envelope structural integration, lightning protection and regulatory classification (hybrid airship vs rotorcraft) are still open. Investors should treat Gen-3B as a second-wave product that benefits from Gen-3A technology maturation.

9. COMPARATIVE SUMMARY OF BOTH PLATFORMS

Attribute	Gen-3A	Gen-3B
Operational mass	100 t	200 t
Primary lift	Pure mechanical	Hybrid (He + mechanical)
Mechanical T/W	≈ 1.45–1.50	≈ 0.45–0.60
Installed power	55–65 MW	70–90 MW
Peripheral fans	6–8 × 11–12 m	8–10 × 11–12 m
Capsule underbody fans	4 × 5–6 m	4–6 × 5.5–6.5 m
Helium volume	None	180–220 × 10^3 m^3
Separation time target	5–7 s	5–7 s
Parachute canopies	6–8	8
Passengers	24–30	28–36
Development priority	First / nearer-term	Follow-on after Gen-3A
Relative technical risk	High but manageable	Higher (buoyancy + scale)

10. SAFETY ARCHITECTURE — CAPSULE SEPARATION IN DETAIL

Separation is the defining safety feature and the highest-risk engineering element. Target sequence (5–7 s total):

1. Threat detection (structural, fire, flight-path) or pilot command + flight-control authorisation.
2. Mechanical and electrical disconnect of all capsule interfaces (power, data, fluid, structural).
3. Separation impulse (pyrotechnic or high-pressure pneumatic) producing a clean relative trajectory.
4. Immediate activation of capsule underbody ducted fans for attitude stabilisation.
5. Sequenced parachute deployment with progressive reefing stages.
6. Controlled descent or splashdown and automatic activation of flotation and distress beacons.

After separation the capsule is fully independent. The 5–7 s window is deliberately longer than earlier 3.5–4 s targets to give realistic time for sensing, decision logic, physical separation and initial canopy inflation, while remaining far faster than any conventional emergency egress from a heavy helicopter.

10.1 Parachute and Post-Separation Performance

6–8 canopies arranged in independent modules with sequenced opening and reefing reduce entanglement risk relative to earlier 12-canopy concepts. Underbody fans provide limited translational and rotational control after separation, allowing attitude maintenance and vertical-speed reduction before full canopy inflation. The combination is sized for a touchdown/splashdown vertical speed compatible with the capsule's crashworthy structure and occupant protection.

10.2 Hydrodynamic Hull and Water Impact

Trimaran/catamaran-inspired planform with crushable central keel and energy-absorbing underside. Geometry limits peak water-impact loads, provides directional stability during water taxi, and supports flotation after hard landing. Detailed hydrodynamic coefficients and slamming loads require model-basin validation. High-voltage systems isolate automatically on water contact; fuel is confined to protected central bladders.

11. SYSTEMS ARCHITECTURE OVERVIEW

11.1 Electrical Power

Distributed high-voltage DC architecture fed by multiple turbogenerators. Bus segmentation and automatic isolation protect against cascading failures. Capsule carries its own battery energy storage sized for the full post-separation mission with reserve. Thermal management of the capsule battery is a critical design driver and remains an open item.

11.2 Flight Control

Fly-by-wire with at least triple redundancy. Control allocation must handle the multi-fan layout, including reconfiguration after loss of one or more peripheral fans and after capsule separation. The separation function is implemented as a high-integrity, independently powered path.

11.3 Modular Interfaces

All major modules (capsule, fan arms, turbogenerator packs) use standardised mechanical, electrical and data interfaces. These enable both rapid assembly during manufacturing and controlled jettison in flight. Interface loads under normal flight, water impact and the separation impulse must be validated by analysis and structural test.

11.4 Avionics and Mission Systems

Modern glass-cockpit philosophy with multiple large displays, synthetic vision and advanced flight-management capability. Mission systems (cargo handling, external load, maritime sensors) are modular. All critical avionics remain inside the safety capsule so they survive separation.

11.5 Manufacturing and Maintainability

Modular architecture supports parallel manufacturing of major assemblies and simplified field maintenance. Fan modules, turbogenerator packs and the capsule itself are designed as line-replaceable units at the appropriate maintenance level. Access panels, lifting points and interface standardisation are to be defined in the next design cycle. No exotic manufacturing processes are required for the baseline concept. Primary composite structure is assumed compatible with out-of-autoclave and automated fibre-placement processes already in industrial use.

12. COMPETITIVE LANDSCAPE

Platform	Mass class	Key limitation vs Gen-3
CH-47 Chinook / CH-53K	~20–25 t	No independent occupant capsule; limited water crashworthiness
Mi-26	~50–56 t	Same architecture limitations; ageing fleet
Civil heavy (AW101 etc.)	~15–20 t	No rapid independent escape system
Existing hybrid airships	Various	Low speed / limited VTOL; no rapid capsule
Gen-3A / Gen-3B concept	100 / 200 t	Unique rapid capsule + amphibious heavy lift

No current production aircraft combines true heavy VTOL, amphibious capability and an independent occupant survival capsule. That gap is the commercial rationale.

13. DEVELOPMENT PATH AND CAPITAL VIEW

13.1 Phased Technical Roadmap

- Phase 0–1 (12–18 months): Parametric freeze, mission definition, preliminary mass/power closure with supplier data, early certification-authority engagement.
- Phase 2–4 (18–30 months): High-fidelity CFD/FEA, aeroelastic assessment, control-law simulation including full separation dynamics, supplier down-select.
- Phase 5–6 (24–36 months): Subscale aero/acoustic tests, model-basin hull trials, instrumented capsule drop tests, multi-canopy deployment trials, battery abuse testing.
- Phase 7–8 (36–60+ months): Unmanned demonstrator focused on hover, water operations and separation; progressive envelope expansion; formal certification-basis development.

13.2 Order-of-Magnitude Capital (Indicative Only)

A first unmanned hover-and-separation demonstrator for a Gen-3A-class vehicle is expected to require capital in the mid-to-high hundreds of millions of USD (airframe, propulsion, systems, testing and a focused core team), depending on technology reuse and demonstrator scale. Full type certification of a manned 100 t aircraft would require a multiple of that figure and a longer calendar. Gen-3B would follow only after Gen-3A technology has been substantially de-risked. These figures are indicative; a formal costed programme plan is a prerequisite for any serious funding discussion.

13.3 Recommended Investment Sequencing

Treat Gen-3A as the primary near-term vehicle and Gen-3B as a follow-on that re-uses capsule, control and modular technology. Early capital should prioritise: (1) closing mass and power budgets with real supplier data, (2) high-fidelity separation simulation plus subscale drop tests, and (3) structured early dialogue with the relevant certification authority on the novel safety architecture.

14. PRINCIPAL RISKS AND MITIGATION DIRECTIONS

Risk	Level	Mitigation Direction
Capsule separation timing & reliability	High	5–7 s window, high-integrity sequencing, extensive drop testing
Multi-fan interference & aeroelastics	Med-High	Modest fan count, spacing, coupled analysis, subscale tests
Installed power mass & availability	Med-High	Conservative T/W, battery buffer, early supplier engagement
Helium envelope (Gen-3B)	Med-High	Multi-cell design, airship practice, Gen-3A first
Certification path & timeline	High	Early authority engagement, clear safety case around capsule
Water ingestion & hull impact loads	Medium	Model-basin programme, fan placement, crushable structure
Parachute entanglement	Medium	Fewer canopies, modular sequenced deployment, reefing
Battery thermal runaway (capsule)	Medium	Cell chemistry, containment, thermal management
Capital intensity & schedule slip	High	Phased funding gates, demonstrator-first, Gen-3A priority
Manufacturing cost & supply chain	Medium	Modular design, parallel assembly, early supplier involvement

All risks remain open at Rev 00-R and require quantitative assessment, ranking and tracking in subsequent phases.

15. REQUIREMENTS AND VERIFICATION MATRIX

ID	Requirement	Method	Status
G3-COM-001	Mechanical T/W ≈ 1.45–1.50 (Gen-3A)	Analysis	Open
G3-CAP-001	Capsule separation in 5–7 s	Sim + Drop test	Open
G3-CAP-002	6–8 canopy sequenced deployment	Sim + Drop test	Open
G3-AMP-001	Water impact / flotation Sea State 3–4	Test + CFD	Open
G3-POW-001	Installed power closes with 15–20 % margin	Analysis	Open
G3-HE-001	Helium multi-cell containment (Gen-3B)	Analysis + Test	Open
G3-AERO-001	Acceptable multi-fan interference	CFD + Test	Open
G3-AERO-002	Aeroelastic stability of fan array	Analysis + Test	Open
G3-SYS-001	Triple-redundant FC + independent sep. path	Analysis + HIL	Open
G3-STR-001	60–70 % composite mass fraction achievable	Analysis + Coupon	Open
G3-MFG-001	Modular LRU manufacturing approach feasible	Analysis	Open

16. OPEN ISSUES — HONEST LIST

- Real turbogenerator specific power, volume, cost and delivery schedule once supplier data are locked.
- Detailed mass of the helium envelope and its structural integration (Gen-3B).
- Quantified multi-fan aerodynamic interference, download and far-field noise.
- High-fidelity separation dynamics simulation followed by full-scale drop-test validation.
- Battery thermal-runaway containment and thermal management inside the capsule.
- Regulatory classification and applicable certification basis for the novel safety architecture.
- Manufacturing cost model and supply-chain capacity for large composite modules and fans.
- Human-factors validation of the capsule interior under normal and emergency conditions.
- Long-term helium leakage rates, inspection intervals and repair procedures.
- Insurance and liability framework for an aircraft that relies on rapid occupant separation.
- Software assurance level and dissimilar-lane strategy for flight-control and separation functions.
- Lightning attachment, current paths and indirect-effects protection for airframe and helium envelope.

17. DESIGN SENSITIVITIES AND TRADE SPACE

Key sensitivities that will be quantified in Phase 0–1 parametric work:

- Fan diameter vs count vs disk loading vs interference vs structural mass of arms.
- Mechanical T/W vs installed power vs fuel burn vs empty-mass fraction.
- Helium volume vs residual mechanical thrust vs envelope structural mass (Gen-3B).
- Capsule mass vs parachute area vs descent rate vs post-separation fan power.
- Battery energy vs post-separation endurance vs capsule mass and thermal load.
- Composite mass fraction vs crashworthiness vs lightning-protection mass.
- Separation timeline vs sensing/decision reliability vs physical clearance margins.

These trades will be used to freeze the Gen-3A configuration before major supplier commitments and before the first high-fidelity separation simulations.

18. MANUFACTURING AND SUPPLY-CHAIN CONSIDERATIONS

The modular architecture is intended to support parallel manufacturing of major assemblies. Primary composite structure is assumed compatible with out-of-autoclave and automated fibre-placement processes already used in large aerospace structures. Fan modules, turbogenerator packs and the capsule are designed as line-replaceable units. Early engagement with turbogenerator, large-fan and battery suppliers is required to lock specific power, volume and cost assumptions. No exotic processes are required for the baseline concept; the principal manufacturing risks are scale, tooling investment and supply-chain capacity for the large composite and rotating components.

19. CERTIFICATION PATH CONSIDERATIONS

The novel safety architecture (rapid independent capsule separation) has no direct precedent in existing rotorcraft certification bases. Early and continuous engagement with the relevant aviation authority is essential. Possible paths include adaptation of existing rotorcraft rules, hybrid-airship precedents (for Gen-3B), or a special-condition / equivalent-safety approach centred on the capsule as the primary means of occupant protection. The safety case must demonstrate that the residual risk after separation is lower than the residual risk of conventional heavy-helicopter crashworthiness. This remains one of the highest programme risks and a primary driver of schedule and cost.

20. KEY INVESTOR TAKE-AWAYS

1. The safety architecture (independent capsule with its own lift and parachutes) is genuinely differentiated and addresses a real residual risk that existing heavy helicopters have never solved.

2. Gen-3A is the rational first product. Its power, mass and separation targets have been set at levels that are still ambitious but more defensible than earlier concept iterations.

3. Gen-3B remains attractive as a capacity step-up but should be funded only after Gen-3A has de-risked the capsule, control laws and modular interfaces.

4. The largest remaining uncertainties are: (a) reliable 5–7 s separation under real conditions, (b) multi-fan aeroelastics and interference, (c) real turbogenerator and battery performance at the required scale, and (d) the certification path for a novel safety system.

5. Capital intensity is high. A phased, demonstrator-first approach with clear technical gates is the only responsible funding strategy.

6. This document is a concept baseline, not a frozen design. Every major quantitative target remains subject to revision as analysis, supplier data and test results become available.

7. Manufacturability has been improved relative to earlier baselines by reducing fan count, retaining conventional industrial processes, and designing major elements as modular line-replaceable units.

21. SUMMARY OF IMPROVEMENTS OVER EARLIER CONCEPT BASELINES

- Mechanical T/W reduced from 1.7 to ≈ 1.45–1.50 → lower power and structural load.
- Separation timeline relaxed from ≤ 3.5–4 s to 5–7 s → more realistic sequencing margin.
- Fan counts reduced → lower interference, simpler control, better manufacturability.
- Parachute count reduced to 6–8 → lower entanglement risk.
- 15–20 % development margins retained in mass and power.
- Gen-3B helium volume sized for genuine 50–65 % buoyancy credit with residual mechanical T/W ≈ 0.45–0.60.
- Installed power budgets lowered into more credible bands (55–65 MW / 70–90 MW).
- Explicit manufacturing and certification considerations added for investor transparency.
- Risk register and open-issues list expanded and written for due-diligence scrutiny.

22. GLOSSARY OF KEY TERMS

CDD — Conceptual Design Document — the present requirements and concept baseline.
CFD — Computational Fluid Dynamics — numerical simulation of fluid flow.
FEA — Finite Element Analysis — numerical structural analysis.
T/W — Thrust-to-Weight ratio — total available thrust divided by vehicle weight.
Disk Loading — Thrust (or weight) divided by total rotor/fan disk area.
Induced Power — Ideal power required to produce thrust in hover (momentum theory).
Capsule — Central, rapidly detachable, independently buoyant occupant compartment.
Jettison — Controlled separation and disposal of a modular component under flight-control authority.
Sea State 3–4 — Moderate sea with significant wave heights typically 0.5–2.5 m.
Multi-cell Envelope — Helium containment subdivided into independent cells to limit single-leak effect.
Reefing — Temporary restriction of parachute canopy opening to limit peak inflation loads.
HIL — Hardware-in-the-Loop — real-time simulation that includes actual control hardware.
LRU — Line-Replaceable Unit — modular component designed for field replacement.
Rev 00-R — Realistic concept baseline revision; not a frozen design.

23. DOCUMENT CONTROL AND APPROVAL STATUS

Item	Detail
Founder / Principal Author	Amirbahador Atai
Document Title	Gen-3 Safe Helicopter — Full Investor & Technical CDD
Revision	00-R Full (Realistic Safety-Centered Baseline)
Date	2026
Status	Concept Baseline / Pre-Feasibility — Full Edition
Classification	Proprietary Concept Information
Engineering Approval	Pending
Certification Status	Not certified — Concept only
Next Revision Trigger	Closure of mass/power budgets with supplier data + first separation simulation results

24. FINAL STATEMENT

This Rev 00-R Full document establishes a coherent, safety-centred, more realistic baseline for both the 100-ton pure rotorcraft (Gen-3A) and the 200-ton hybrid buoyant rotorcraft (Gen-3B). The central safety capsule with underbody fans and rapid multi-canopy separation remains the defining feature of the Gen-3 family. All quantitative targets are provisional and must be confirmed by the analysis and test programme outlined in the validation roadmap.

The document has been written to be useful both to technical reviewers and to investors conducting due diligence. It deliberately trades some of the more aggressive earlier performance claims for improved credibility, manufacturability and a clearer path to a first unmanned demonstrator. Gen-3A is positioned as the nearer-term product; Gen-3B as a capacity follow-on that re-uses the core safety and modular technology.

Subsequent revisions should close the mass and power budgets with real supplier data, present the first quantified separation-simulation results, and refine the capital and schedule estimates on the basis of a formal costed programme plan.

End of Document — Gen-3 Safe Helicopter / Heavy Modular Amphibious VTOL Platform
Conceptual Design Document Rev 00-R Full (Realistic Investor & Technical Baseline) — 2026

This page intentionally contains closing remarks and document control continuation to complete the 16-page structure. All technical content, tables, assumptions, risks and investor guidance are contained in the preceding sections. Readers seeking the quantitative baselines should refer to Sections 7 (Gen-3A), 8 (Gen-3B), 9 (Comparison), 14 (Risks) and 20 (Investor Take-aways).

Contact for authorised discussion of this concept baseline: via the Founder, Amirbahador Atai. Any distribution beyond authorised recipients requires prior written consent.

25. APPENDIX A — QUICK REFERENCE PERFORMANCE CARD

Gen-3A at a Glance

Item	Value
Max operational mass	100 t
Mechanical T/W target	1.45–1.50
Installed power	55–65 MW
Peripheral + capsule fans	6–8 × 11–12 m + 4 × 5–6 m
Separation time	5–7 s
Parachutes	6–8 modular canopies
Passengers	24–30 + 2 crew
Development priority	First product

Gen-3B at a Glance

Item	Value
Max operational mass	200 t
Helium volume	180–220 × 10^3 m^3
Buoyancy credit	~50–65 % of mass
Mechanical T/W residual	0.45–0.60
Installed power	70–90 MW
Peripheral + capsule fans	8–10 × 11–12 m + 4–6 × 5.5–6.5 m
Separation time	5–7 s
Parachutes	8 modular canopies
Passengers	28–36 + 2 crew
Development priority	Follow-on after Gen-3A

These quick-reference cards summarise the frozen conceptual targets of Rev 00-R Full. They are intended for rapid orientation by investors, partners and technical reviewers. Full context, assumptions, risks and open issues are contained in the main body of the document.

End of Appendix — Gen-3 Safe Helicopter CDD Rev 00-R Full — 2026